

TEST REPORT

Production AI Anomaly Detection & Network Forensics Evaluation • Analyst: Security Analyst (nandini)

TOTAL FLOWS INSPECTED	THREATS DETECTED	CRITICAL INCURSIONS	BENIGN FLOWS	AVERAGE RISK
1,727	855	473	872	44.4/100

1. PRODUCTION MACHINE LEARNING MODEL PERFORMANCE

Model Architecture	Accuracy	Precision	Recall	F1 Score	Pipeline Features	Status
Random Forest Classifier	100.0%	100.0%	100.0%	100.0%	78 Flow Metrics	ACTIVE INFERENCE

2. MONITORED ATTACK CLASSIFICATIONS & RISK SCORING

Classification	Detected Flows	Traffic Share	Severity	Risk Rating	Detection Signature
BENIGN	872	50.5%	LOW	5/100	Normal Baseline Traffic
DDoS	473	27.4%	CRITICAL	95/100	High-Rate Ingress Syn Flood
FTP-Patator	210	12.2%	MEDIUM	65/100	Password Spray Incursion
SSH-Patator	172	10.0%	HIGH	80/100	Brute-Force SSH Auth Attempt

3. RECENT SECURITY INCURSIONS & HOST TELEMETRY

Threat ID	Attack Type	Source IP	Destination Target	Confidence	Severity
#855	DDoS	172.16.114.38	10.0.0.14:TCP	100.0%	CRITICAL
#854	SSH-Patator	192.168.12.133	10.0.0.8:TCP	84.0%	HIGH
#853	DDoS	172.16.171.124	10.0.0.16:TCP	100.0%	CRITICAL
#852	FTP-Patator	192.168.10.104	10.0.0.14:TCP	64.0%	MEDIUM
#851	SSH-Patator	192.168.12.131	10.0.0.15:TCP	94.0%	HIGH

INCIDENT RESPONSE PROTOCOL:

- **Automated Firewall Containment:** Enforce rate limiting on external IP ranges exhibiting DDoS patterns.
- **Authentication Security:** Apply temporary IP blocks on hosts generating repetitive SSH/FTP auth failures.
- **Perimeter Inspection:** Ensure real-time Snort/Suricata and NetShield AI heuristic rule sync is active across all ingress gateways.

